

BIOL-1 Practice Test 05

Module 9: Inheritance Genetics

Instructions: This practice test covers material from Module 9. Answer all questions to the best of your ability.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

1. Gregor Mendel studied inheritance patterns using:

- A) Fruit flies
- B) Mice
- C) Pea plants
- D) Bacteria

2. An organism with the genotype "Aa" is:

- A) Homozygous dominant
- B) Homozygous recessive
- C) Heterozygous
- D) Codominant

3. If brown eyes (B) are dominant to blue eyes (b), what is the phenotype of "Bb"?

- A) Blue eyes
- B) Brown eyes
- C) A mix of brown and blue
- D) Green eyes

4. When you cross $Tt \times Tt$, the phenotypic ratio is:

- A) 1:1
- B) 2:1
- C) 3:1
- D) 4:0

5. A test cross involves crossing an individual with a dominant phenotype with:

- A) Another dominant individual
- B) A homozygous recessive individual
- C) A heterozygous individual
- D) Itself

6. The expected phenotypic ratio from a dihybrid cross ($AaBb \times AaBb$) is:

- A) 3:1
- B) 1:2:1
- C) 9:3:3:1
- D) 1:1:1:1

7. In incomplete dominance, what does the heterozygote look like?

- A) Like the dominant parent
- B) Like the recessive parent
- C) A blend of both parents
- D) Neither parent

8. Red snapdragons (RR) crossed with white snapdragons (rr) produce pink offspring (Rr).

This is an example of:

- A) Complete dominance
- B) Incomplete dominance
- C) Codominance
- D) Sex-linked inheritance

9. In codominance, the heterozygote shows:

- A) Only the dominant trait
- B) A blend of traits
- C) Both traits fully
- D) Neither trait

10. A person with Type AB blood has the genotype:

- A) $I^A I^A$
- B) $I^B I^B$
- C) $I^A I^B$
- D) ii

11. A person with Type O blood has the genotype:

- A) $I^A i$
- B) $I^B i$
- C) $I^A I^B$
- D) ii

12. Can two parents with Type A and Type B blood have a Type O child?

- A) Never
- B) Yes, if both parents are heterozygous
- C) Only if the mother is Type A
- D) Only if the father is Type B

13. Color blindness is an X-linked recessive trait. Which person is most likely to be colorblind?

- A) A female with one carrier allele
- B) A male with one recessive allele on his X
- C) A female with two dominant alleles
- D) A male with a dominant allele on his X

14. Can a colorblind father pass color blindness to his sons?

- A) Yes, always
- B) Yes, sometimes
- C) No, never (fathers give sons the Y chromosome)
- D) Only if the mother is also colorblind

15. Two unaffected parents have a child with a genetic disorder. The trait is most likely:

- A) Dominant
- B) Recessive
- C) X-linked dominant
- D) Polygenic

16. In a pedigree, a circle represents:

- A) A male
- B) A female
- C) An affected individual
- D) A carrier

17. A polygenic trait is controlled by:

- A) One gene
- B) Two alleles
- C) Many genes
- D) The environment only

18. Human height is an example of:

- A) Simple dominant/recessive inheritance
- B) Codominance
- C) Polygenic inheritance
- D) Sex-linked inheritance

19. Mendel's Law of Segregation states that:

- A) Genes on different chromosomes sort independently
- B) Alleles separate during gamete formation

- C) Dominant alleles always win
- D) All genes are linked

20. Mendel's Law of Independent Assortment applies to genes that are:

- A) On the same chromosome
- B) On different chromosomes
- C) Linked together
- D) Sex-linked

21. What is a carrier?

- A) Someone who is homozygous dominant
- B) Someone who is heterozygous for a recessive trait
- C) Someone who is affected by a disorder
- D) Someone with a polygenic trait

22. Sickle cell anemia is an example of pleiotropy because:

- A) It involves many genes
- B) One allele affects multiple traits (blood, spleen, oxygen)
- C) It is sex-linked
- D) It shows codominance

23. If skin color is controlled by three genes, the phenotypic distribution would be:

- A) Two distinct categories
- B) Three categories
- C) A continuous range (bell curve)
- D) Completely random

24. Hydrangea flowers change color based on soil pH. This shows that:

- A) Genotype completely determines phenotype
- B) Environment can influence phenotype
- C) Flowers have no genotype
- D) Soil pH changes DNA

25. The F₂ generation is the offspring of:

- A) The original parents (P₀)
 - B) The F₁ generation crossed with each other
 - C) A test cross
 - D) A dihybrid cross only
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Part B: Fill in the Blank (10 questions)

26. The physical appearance of an organism is its _____.

27. The combination of alleles an organism carries is its _____.

28. An organism with two identical alleles (AA or aa) is _____.

29. An organism with two different alleles (Aa) is _____.

30. The expected phenotypic ratio of a monohybrid cross (Aa × Aa) is _____.

31. In the ABO blood type system, Type O blood has the genotype _____.

32. Males are more likely to have X-linked recessive disorders because they only have _____ X chromosome(s).

33. A trait controlled by many genes that shows continuous variation is called a _____ trait.

34. A family tree used to track inheritance of a trait across generations is called a _____.

35. Mendel's Law of _____ states that alleles separate during gamete formation.

Part C: Short Answer (4 questions)

36. Cross a heterozygous tall plant (Tt) with a homozygous short plant (tt). Show the Punnett square and give the phenotypic ratio.

37. Explain the ABO blood type system. Can a Type A mother ($I^A i$) and a Type B father ($I^B i$) have a Type O child? Show your work.

38. Explain why X-linked recessive disorders are more common in males than females.

39. What is the difference between a polygenic trait and a single-gene Mendelian trait? Give an example of each.

End of Practice Test 05